Dual Monte Carlo Uncertainty Framework for Neural Networks: Application to a Quantum Voltage Standard

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***Abstract*—We present a new method compliant with the Guide to the expression of uncertainty in measurement (GUM), the dual Monte Carlo uncertainty framework (DMCUF) for neural networks, which jointly quantifies input measurement uncertainty and neural network model uncertainty. The probability distributions of the inputs are propagated using the GUM Monte Carlo method in an outer loop, while neural network variability is accounted via Monte Carlo Dropout in an inner loop. By combining these outputs we obtain an empirical predictive distribution from which coverage intervals and uncertainty budgets are derived. This approach preserves metrological traceability by mapping each uncertainty source to GUM Type A or B classifications and reporting expanded uncertainties. We validate the DMCUF on three different neural network models trained on DC voltage measurements using a programmable Josephson voltage standard.**

***Index Terms*—Josephson effect, metrology, Monte Carlo methods, neural networks, uncertainty.**

I. Introduction

Quantum voltage standards such as the Programmable Josephson Voltage Standard (PJVS) [1] are the basis for traceable DC voltage realization with extremely low uncertainty. Recently, data-driven models and digital twins have been proposed to complement physics-based descriptions of such systems, enabling fast prediction, optimization, and diagnostic capabilities. Neural networks are particularly attractive due to their ability to approximate nonlinear relationships between operating parameters and measured voltages.

Despite their predictive power, neural networks introduce additional sources of uncertainty that are not explicitly addressed by classical uncertainty budgets. In particular, uncertainty arises both from the input quantities—such as calibration data, environmental effects, and measurement noise—and from the model itself, due to limited training data and imperfect representation of the underlying physical system. For metrological applications, these contributions must be quantified and reported in compliance with the GUM [2].

This work proposes the DMCUF, that combines GUM-compliant Monte Carlo [3] of measurements (input) uncertainties with Monte Carlo Dropout [4] as a practical approximation of neural network uncertainty. The method is demonstrated using DC voltage measurements from a PJVS, part of a neural-network-based digital twin, developed in the scope of the EURAMET ViDiT project [5]. The focus of this paper is on the methodology, rather than on optimizing predictive performance.

II. Method

A. Dual Monte Carlo Uncertainty Framework

Let $\hat{y}$ denote the predictions of a trained neural network represented as

$$\hat{y} = f_\omega(x; m) + b, \tag{1}$$

where x is the vector of input quantities, ω denotes a fixed snapshot of trained weights, m is a stochastic dropout mask, and $b = \overline{\hat{y}_{\mathrm{ref}} - y_{\mathrm{ref}}}$ the bias of model's predictions obtained as the mean of the residuals between reference measurements y_{ref} and model's predictions $\hat{y}_{\mathrm{ref}}$. Input quantities are characterized by calibrated probability density functions (PDFs), including correlations where applicable, and classified as Type A or Type B following the GUM.

Uncertainty propagation is performed using a nested Monte Carlo scheme. In the outer loop, N input samples $x_i \sim p(x)$ are drawn from the joint input PDF. For each x_i, an inner loop performs K stochastic forward passes through the network using Monte Carlo Dropout, producing

$$y_{i,j} = f_\omega(x_i; m_j) + b, \quad j = 1, \ldots, K, \tag{2}$$

where each m_j corresponds to an independent dropout realization. The resulting ensemble $\{y_{i,j}\}$ empirically approximates the predictive distribution of the model, and for each x_i we can evaluate the mean output $\bar{y}_i$.

The total variance of Y is decomposed using the law of total variance,

$$\mathrm{Var}(Y) = \mathbb{E}_X[\mathrm{Var}(Y \mid X)] + \mathrm{Var}_X(\mathbb{E}[Y \mid X]), \tag{3}$$

allowing to evaluate the model-induced and the input-induced uncertainty contributions. These components are estimated from samples as

$$u_{\mathrm{model}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\frac{1}{K-1}\sum_{j=1}^{K}(y_{i,j} - \bar{y}_i)^2\right)}, \tag{4}$$

$$u_{\mathrm{input}} = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(\bar{y}_i - \hat{\mu})^2}, \tag{5}$$

where $\hat{\mu}$ is the overall ensemble mean. Systematic model bias uncertainty u_b is evaluated from reference measurements as a Type A contribution (modelled as normal distribution) and added in quadrature. The combined statistical standard uncertainty is

$$u_c = \sqrt{u_{\text{input}}^2 + u_{\text{model}}^2 + u_b^2}. \quad (6)$$

Monte Carlo Dropout is interpreted as a variational approximation to the Bayesian posterior over network weights [4], and the associated model variability is treated as a Type A contribution estimated by repeated sampling. Expanded uncertainties or percentile-based coverage intervals are then derived directly from the empirical predictive distribution, consistent with GUM Supplement 1 recommendations [3].

III. Results

The framework was evaluated using DC voltage measurements, generating a 1 V step in the 10 V PJVS systems of TÜBİTAK UME and INTI. Three neural network architectures were tested as surrogate models within the PJVS digital twin to predict the DC voltage steps: a multilayer perceptron (MLP) as a feedforward baseline, a long short-term memory network (LSTM) that captures temporal dependences, and a convolutional neural network (CNN) that extract local patterns. All models were trained on the same temporally-split dataset (70% training, 20% validation, 10% test) in a one-step-ahead (horizon = 1) sequence-to-sequence setting. The DMCUF outer loop used $N = 1000$ input samples (GUM Monte Carlo) with an input standard uncertainty estimated from the measurement series ($\sigma_{\text{input}} = 79\,\text{nV}$), while the inner loop used $K = 200$ Monte Carlo Dropout forward passes (dropout rate = 40%). Reported uncertainties are standard uncertainties (coverage factor $k = 1$) and 95% empirical coverage is approximated by the interval mean $\pm\, 2u_c$.

A. Predictive accuracy and bias

All three architectures achieved comparable point-prediction accuracy on the test set. The root-mean-square error (RMSE) is on the order of 70 nV. The mean predictive bias for all models is small relative to the RMSE ($2 - 20$ nV), capturing the principal deterministic structure of the series generated by the PJVS system with similar effectiveness.

Table I summarizes the metrics: RMSE (used only to evaluate the quality of predictions), mean bias, and the empirical coverage of the approximate 95% interval (mean $\pm\, 2u_c$).

TABLE I
Comparison of predictive performance (test set).

Model	RMSE (nV)	Bias (nV)	Coverage 95%
MLP	64	-2	96.47%
LSTM	66	-13	90.93%
CNN	69	-17	73.05%

B. Uncertainty decomposition

The uncertainty decomposition into the mean input contribution $\langle u_{\text{input}} \rangle$, the mean model contribution $\langle u_{\text{model}} \rangle$, the bias contribution u_b, and the mean combined standard uncertainty $\langle u_{\text{c}} \rangle$ is reported in Table II.

TABLE II
Comparison of uncertainty components (test set).

Model	u_b (nV)	$\langle u_{\text{input}} \rangle$ (nV)	$\langle u_{\text{model}} \rangle$ (nV)	$\langle u_{\text{c}} \rangle$ (nV)
MLP	3	22	67	71
LSTM	3	16	50	53
CNN	3	9	33	35

For the MLP and the LSTM we observed that both contributions are of the same order of magnitude and that their quadrature sum produces an uncertainty band that mostly covers the observed test dispersion; empirical 95% coverage reached 96.47% and 90.93%, respectively. This indicates that the stochastic contributions captured by DMCUF explain a substantial fraction of the test variability.

The CNN model exhibits smaller reported uncertainty that underestimates the test dispersion (73.05%). This results from a combination of relatively low sensitivity of the CNN outputs to the particular input perturbation scale used, and a smaller dispersion across MC-Dropout realizations for the chosen dropout configuration. Possible solutions evaluated include increasing dropout or MC sampling (larger K), or combining MC-Dropout with a small deep ensemble; alternatively, a validation-derived variance calibration factor α may be applied and reported as a Type-B correction.

IV. Conclusion

A GUM-compliant dual Monte Carlo uncertainty framework has been presented for neural-network-based models and demonstrated on a PJVS application. By combining Monte Carlo input propagation with Monte Carlo Dropout, the method provides a transparent and reproducible way to quantify uncertainty contributions from both inputs and model structure. The approach is readily applicable to other metrological systems employing data-driven models.

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